

MA511: Computer Programming

Practice Lab 1 (Due: 19th Aug 2025)

To execute these labs, you will only use the tools/techniques we have discussed in class.

1. Write a C program to convert Fahrenheit to Celsius using macros. The formula is $(\text{<Farhenheit>} - 32) \times 5/9 = \text{<Celsius value>}$.
2. Write a program to swap two numbers using macors.
3. Write a function to compute the factorial of an integral number. The program should be interactive. Use the do – while loop. A sample output is shown. The user inputs are marked in red.

Do you want to compute factorial (y/n)? **y**

Please enter the number: **4**

The factorial of 4 is 24.

Do you want to continue(y/n)?**y**

Please enter the number: **3**

The factorial of 3 is 6.

Do you want to continue(y/n)?**n**

4. Write a macro `DEBUG(msg)` that prints the filename, line number, and a custom message.
5. Write a macro `POWER(x, y)` that expands to `pow(x, y)`. Also implement your own function `int power(int base, int exp)` using loops. Compare results.

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